#A Check the current ETag matches the one in the request.
#B If not, return a 412 Precondition Failed response.
#C Otherwise return the new ETag after updating the resource.

The ETag mechanism can also be used to prevent replay of a PUT request that is intended to create a resource that doesn't yet exist. Because the resource doesn't exist, there is no existing ETag or Last-Modified date to include. An attacker could replay this message to overwrite a later version of the resource with the original content. To prevent this, you can include an If-None-Match header with the special value *, which tells the server to reject the request if there is any existing version of this resource at all.

> **TIP** The *Constrained Application Protocol* (CoAP), often used for implementing REST APIs in constrained environments, doesn't support the Last-Modified or If-Unmodified-Since headers, but it does support ETags along with If-Matches and If-None-Match. In CoAP, headers are known as *options*.

ENCODING HEADERS WITH END-TO-END SECURITY

As explained in chapter 12, in an end-to-end IoT application, a device may not be able to directly talk to the API in HTTP (or CoAP) but must instead pass an authenticated message through multiple intermediate proxies. Even if each proxy supports HTTP, the client may not trust those proxies not to interfere with the message if there is not an end-to-end TLS